

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1401

Walkersville 138 kV
September 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1401. The Interconnection Customer proposes a 115.5 MW solar photovoltaic facility interconnecting at the Walkersville 138 kV substation in the Maryland area. The request was received on February 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 423 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,322,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ROSSMOOR S to TRUMANN 138 kV L
Contingency: TIGER BAY SUBSTATION to AIR FORCE BASE 345 kV
Loading: 177.2% (Rating: 229 MVA)
DFAX: 6.1%
Pre-project loading: 174.2%
2. Facility: ROSSMOOR S to TRAMWAY 138 kV
Contingency: TIGER BAY to AIR FORCE BASE 345 kV
Loading: 147.5% (Rating: 313 MVA)
DFAX: 6.1%
Pre-project loading: 145.3%
3. Facility: BONDURANT to MONTGOMERY VILLAGE SUBSTATION 138 kV Circuit 1
Contingency: MONTGOMERY VILLAGE to BONDURANT 138 kV
Loading: 146.8% (Rating: 361 MVA)
DFAX: 6.3%
Pre-project loading: 144.8%
4. Facility: BONDURANT to TRAMWAY 138 kV Circuit 1
Contingency: TRAMWAY to BONDURANT 138 kV Cir. 2
Loading: 142.2% (Rating: 360 MVA)
DFAX: 6.3%
Pre-project loading: 140.1%
5. Facility: BONDURANT to TRAMWAY 138 kV Cir. 2
Contingency: BONDURANT to TRAMWAY 138 kV
Loading: 138.8% (Rating: 368 MVA)
DFAX: 6.3%
Pre-project loading: 136.9%

6. Facility: MONTGOMERY VILLAGE to BONDURANT 138 kV Ckt 2
Contingency: MONTGOMERY VILLAGE to BONDURANT 138 kV Line 1
Loading: 134.2% (Rating: 395 MVA)
DFAX: 6.3%
Pre-project loading: 132.3%
7. Facility: MONTGOMERY VILLAGE to AIR FORCE BSE 138 kV #1
Contingency: AIR FORCE BASE to MONTGOMERY VILLAGE 138 kV
Loading: 133.9% (Rating: 372 MVA)
DFAX: 5.0%
Pre-project loading: 132.3%
8. Facility: G-P GYPSUM SUBSTATION to CLOVER VLY 138 kV
Contingency: ROSS FORK to G-P GYPSUM SUBSTATION 138 kV L
Loading: 127.3% (Rating: 161 MVA)
DFAX: 5.1%
Pre-project loading: 123.7%
9. Facility: MONTGOMERY VILLAGE to AIR FORCE BASE 138 kV Circuit 2
Contingency: AIR FORCE BASE SUBSTATION to MONTGOMERY VILLAGE 138 kV Line 1
Loading: 125.4% (Rating: 397 MVA)
DFAX: 5.0%
Pre-project loading: 124.0%
10. Facility: CLOVER VLY to MIDDLEBROOK 138 kV
Contingency: ARTESIA COUNTRY CLUB to AIR FORCE BASE 345 kV
Loading: 120.2% (Rating: 204 MVA)
DFAX: 7.7%
Pre-project loading: 115.9%
11. Facility: MCCAIN to LAMONT 138 kV
Contingency: CLARK COUNTY to ARTESIA COUNTRY CLUB 345 kV
Loading: 118.9% (Rating: 205 MVA)
DFAX: 7.3%
Pre-project loading: 114.8%
12. Facility: CLARK COUNTY to MCCAIN 138 kV
Contingency: ARTESIA COUNTRY CLUB SUBSTATION to CLARK COUNTY 345 kV L
Loading: 117.6% (Rating: 219 MVA)
DFAX: 7.3%
Pre-project loading: 113.8%
13. Facility: AIR FORCE BASE to RANCHO 138 kV Line
Contingency: ARTESIA COUNTRY CLUB to AIR FORCE BASE 345 kV
Loading: 102.6% (Rating: 269 MVA)
DFAX: 7.8%
Pre-project loading: 99.3%
14. Facility: MIDDLEBROOK to RANCHO 138 kV
Contingency: AIR FORCE BASE to ARTESIA COUNTRY CLUB 345 kV
Loading: 101.5% (Rating: 296 MVA)
DFAX: 7.8%
Pre-project loading: 98.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROSSMOOR S to TRUMANN 138 kV L	Rebuild line with higher-capacity conductor	\$4,186,000	81% (\$3,398,000)
ROSSMOOR S to TRAMWAY 138 kV	Rebuild line with higher-capacity conductor	\$2,882,000	81% (\$2,340,000)
MONTGOMRY VILLAGE SUBSTATION to BONURANT 138 kV #1	Rebuild line with higher-capacity conductor	\$5,537,000	100% (\$5,537,000)
BONDURANT to TRAMWAY 138 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,939,000	100% (\$1,939,000)
BONDURANT to TRAMWAY 138 kV	Rebuild line with higher-capacity conductor	\$2,071,000	100% (\$2,071,000)
MONTGOMERY VILLAGE to BONDURANT 138 kV line	Rebuild line with higher-capacity conductor	\$5,503,000	100% (\$5,503,000)
MONTGOMERY VILLAGE to AIR FORCE BSE 138 kV #1	Rebuild line with higher-capacity conductor	\$872,000	100% (\$872,000)
G-P GYPSUM SUBSTATION to CLOVER VLY 138 kV	Rebuild line with higher-capacity conductor	\$11,836,000	100% (\$11,836,000)
MONTGOMERY VILLAGE to AIR FORCE BASE 138 kV Circuit 2	Rebuild line with higher-capacity conductor	\$595,000	100% (\$595,000)
CLOVER VLY to MIDDLEBROOK 138 kV	Rebuild line with higher-capacity conductor	\$19,676,000	100% (\$19,676,000)
MCCAIN to LAMONT 138 kV	Rebuild line with higher-capacity conductor	\$15,763,000	33% (\$5,158,000)
MCCAIN to CLARK COUNTY 138 kV	Rebuild line with higher-capacity conductor	\$3,131,000	33% (\$1,024,000)
AIR FORCE BASE to RANCHO 138 kV Line	Reconductor line	\$238,000	50% (\$119,000)
RANCHO to MIDDLEBROOK 138 kV	Reconductor line	\$508,000	50% (\$254,000)
	Total allocated to project		\$60,322,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.